App Dev Project Report

1. Student Details

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About Me: I am a student of the IIT Madras BS Degree program with a strong interest in web development, artificial intelligence, and machine learning. I enjoy building practical applications that combine technology, problem-solving, and user experience while continuously developing my technical skills.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

To design and build a web-based application that enables admins, trek staff, and trekkers to efficiently manage trekking activities, staff assignments, bookings, and trek history.

Approach:

The application was built using Flask, Jinja2, Bootstrap, and SQLite with role-based access. It allows admins to manage treks and staff, staff to manage assigned treks and participants, and trekkers to search, book, and track their trekking activities.

3. AI/LLM Declaration

I used ChatGPT (GPT-5) as a supplementary tool during the development of my MAD-1 project, primarily for UI/UX ideas, layout and styling suggestions, Flask/SQLAlchemy clarification, debugging, and documentation support.

The overall AI/LLM usage was approximately 10–15%, with most assistance focused on UI/UX and presentation. AI suggestions were reviewed and adapted by me, while the application structure, database design, authentication, CRUD and booking logic, integration, testing, and final implementation were developed and verified by me.

4. Description

The project is a Trekking Management System developed using Flask, designed to manage treks, users, trek staff, and bookings through role-based access. The system provides separate functionality for administrators, trek staff, and trekkers, with database-backed CRUD operations, authentication, and booking management. The objective is to provide a structured and user-friendly platform for managing the complete trekking workflow.

5. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
Flask-SQLAlchemy	Database integration with Flask
SQLAlchemy	ORM and database operations

Technology / Library	Purpose
Jinja2	Dynamic HTML page rendering
Bootstrap 5	Responsive UI styling
HTML5	Webpage structure
CSS3	Custom styling and layout
Flask-Login	Authentication and session management
SQLite	Database storage
Python	Backend programming and application logic
Flash Messaging	User action feedback

6. Database Schema / ER Diagram

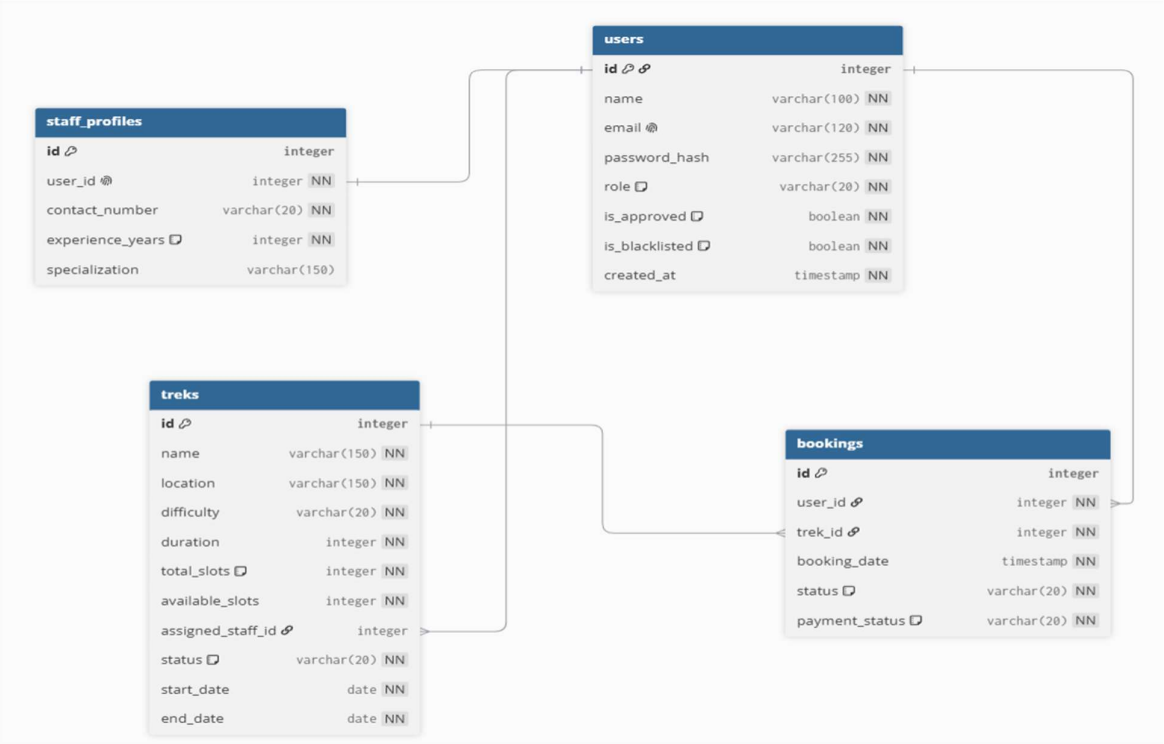
Tables

1. **User** — stores admin, trek staff, and trekker details such as ID, name, email, password, role, approval, and blacklist status.
2. **Trek** — stores trek details including name, location, difficulty, duration, slots, assigned staff, status, and dates.
3. **Booking** — stores trek booking details including user, trek, booking date, booking status, and payment status.
4. **StaffProfile** — stores additional trek staff information such as contact number, experience, and specialization.

Relationships

- **One-to-Many** → User → Trek (assigned treks)
- **One-to-Many** → User → Booking
- **One-to-Many** → Trek → Booking

- **One-to-One** → User → StaffProfile



7. Architecture and Features

Architecture Overview:

- **app.py** – Main Flask application, routing, authentication, and request handling
- **models.py** – SQLAlchemy database models and relationships
- **/templates** – Jinja2 HTML templates for dynamic page rendering
- **/static** – CSS, Bootstrap, and other static resources
- **SQLite Database** – Persistent storage managed through SQLAlchemy
- **Flask-Login** – User authentication, sessions, and role-based access

Implemented Features:

- User registration and login
- Role-based access for Admin, Trek Staff, and Trekker
- Admin trek management and CRUD operations
- Trek staff management and trek assignment
- Trek browsing and detailed trek information
- Trek booking and booking management
- Database relationships using SQLAlchemy

- Form validation and flash messages
- Responsive UI using Bootstrap
- Protected routes and session-based authentication

8. Video Presentation

DriveLink:
https://drive.google.com/file/d/1EQFfhJNlurCPDz3x6yRQCx0xqMZwTDN-/view?usp=sharing